Lesson 25 Counting Functions

In Lesson 23, we saw that there are no one-to-one functions from a finite set A to a finite set B if $|A| > |B|$, this is the Pigeonhole Principle. In this lesson, we discuss more generally the number of functions from A to B. If $|A| = n$ and $|B| = m$, how many functions $f: A \to B$ are there?

A function from A to B is a rule that assigns every element in A to exactly one element in B. How many possible rules are there? Let's look at an example.

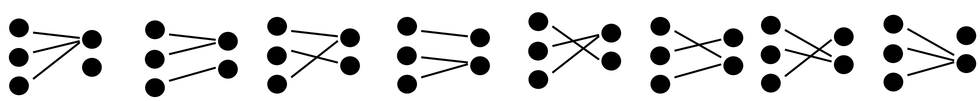
Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. A function rule is defined when we fill in all of the following blanks:

$$f(1) = \_\_\_\_\_\_ \qquad f(2) = \_\_\_\_\_\_\_ \qquad f(3) = \_\_\_\_\_\_\_\_$$

There are two ways to fill each blank: " 4 " or " 5 ". According to the Multiplication Principle, the number ways to fill the 3 blanks is:

$$2 \times 2 \times 2 = 2^3 = 8$$

There 8 possible functions:



In general, each of the n elements in A has m choices where they can go, so:

- The number of functions $f: A \to B$ is

$$\underbrace{m \times m \times \cdots \times m}_{n} = m^n$$

Our second question is the number of one-to-one functions. The Pigeonhole Principle says that if $|A| > |B|$, there are no one-to-one $f: A \to B$ functions. On the other hand, if $|A| \leq |B|$, the first element of A may be mapped to any of the m elements in B, and the second element of A may be mapped to one of the remaining $m - 1$ elements in B, etc.

- If $n \leq m$, the number of one-to-one functions from A to B is:

$$m(m-1)(m-2)\cdots(m-n+1) = P(m, n)$$

If $A = \{1, 2\}$ and $B = \{3, 4, 5, 6\}$, then there are $P(4,2) = 4 \times 3 = 12$ one-to-one functions from A to B.

Finally, we are interested in the number of onto functions. Clearly, if $|A| < |B|$, we have no onto functions, as there are not enough elements in A for each of an element in B to be mapped to. If $|A| \geq |B|$, we may use the inclusion–exclusion principle (Lesson 24) to find the number of onto functions, as explained below.

- Let $B = \{b_1, b_2, \ldots, b_m\}$ and B_i be the set of functions $f: A \to B$ such that b_i is NOT hit. Then the union $B_1 \cup \ldots \cup B_m$ contains exactly all functions which are NOT onto.

- The number of functions in B_i is $(m-1)^n$ because the domain of these functions has n elements while the codomain has $m-1$ elements.
- The intersection $B_i \cap B_j$ is the set of functions that do not map to b_i and b_j, so $B_i \cap B_j$ contains exactly $(m-2)^n$ functions. An intersection of k sets has $(m-k)^n$ functions.
- Since there are m sets: $B_1, B_2, \dots, B_m$, there are $\binom{m}{k}$ intersections of k sets.
- The number is of non-onto functions is then,

$$|B_1 \cup B_2 \cup \dots \cup B_m| = \sum |B_i| - \sum |B_i \cap B_j| + \sum |B_i \cap B_j \cap B_k| + \cdots + (-1)^{m+1} |B_1 \cap B_2 \cap \dots \cap B_m|$$

$$= \binom{m}{1}(m-1)^n - \binom{m}{2}(m-2)^n + \binom{m}{3}(m-3)^n + \cdots + (-1)^m \binom{m}{m-1} 1^n + \cancel{(-1)^{m+1}\binom{m}{m} 0^n}$$

The total number of functions is m^n, so the number of onto functions is:

$$\begin{aligned} m^n - |B_1 \cup B_2 \cup \dots \cup B_m| &= m^n - \left[\binom{m}{1}(m-1)^n - \binom{m}{2}(m-2)^n + \binom{m}{3}(m-3)^n + \cdots + (-1)^m \binom{m}{m-1}\right] \\ &= m^n - \binom{m}{1}(m-1)^n + \binom{m}{2}(m-2)^n - \binom{m}{3}(m-3)^n + \cdots + (-1)^{m-1}\binom{m}{m-1} \end{aligned}$$

In Σ summation notation,

- The number of onto functions from A to B is:

$$\sum_{i=0}^{m-1} (-1)^i \binom{m}{i} (m-i)^n$$

If $|A| = 3$ and $|B| = 2$, how many functions from A to B are onto?

$$\begin{aligned} \sum_{i=0}^{1} (-1)^i \binom{2}{i} (2-i)^3 &= (-1)^0 \binom{2}{0}(2-0)^3 + (-1)^1 \binom{2}{1}(2-1)^3 \\ &= 2^3 - 2(1) = 6 \end{aligned}$$

Finally, since there are no one-to-one functions when $n > m$ and there are no onto functions when $n < m$, there are no bijections unless $n = m$. Then the one-to-one functions are also onto, and they are bijections.

- When $n = m$, there are $m(m-1) \cdots 3 \cdot 2 \cdot 1 = m!$ bijections.